Abhay Mall

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Summary

Aspiring Software Engineer with strong foundations in **C/C++, Python, JavaScript, and Data Structures & Algorithms**. Experienced in developing **backend services, REST APIs, full-stack applications, authentication systems, and database-driven solutions** using Node.js, Express.js, MongoDB, MySQL, and Redis. Solved **multiple coding problems across multiple platforms**, demonstrating strong problem-solving and algorithmic skills. Hands-on experience with **MERN stack, JWT authentication, LLM applications, RAG, embeddings, vector search, and API development**, with a focus on clean, efficient, and maintainable code.

Education

2023 - Present **Bachelor of Engineering (B.E.) in Computer Science and Engineering**
Chandigarh University, Mohali (CGPA: 8.27/10)

2021 **Intermediate (Class 12th)**
Mahatma Gandhi Inter College, Gorakhpur (CGPA: 7.79/10)

2019 **High School (Class 10th)**
Mahatma Gandhi Inter College, Gorakhpur (CGPA: 9.11/10)

Projects

Document Intelligence RAG Assistant **2026**
Technologies: Python, FastAPI, LangChain, Gemini API, FAISS, Embeddings

- Built a **RAG-based document assistant** that processes uploaded documents and answers user questions using retrieved document context.
- Implemented document **chunking, embeddings, vector storage, and similarity search** with FAISS to retrieve relevant information for LLM responses.
- Developed **FastAPI REST endpoints** for document upload and question answering, with prompt-based context injection to reduce irrelevant responses.

AI-Enabled Internship Project Allocation System **2026**
Technologies: MongoDB, Express.js, React.js, Node.js, JWT, REST APIs

- Developed a **MERN-based platform** with modular backend APIs, MongoDB database integration, and secure JWT-based authentication.
- Implemented **role-based dashboards, resume/skill profiles, opportunity management, application workflows, and skill-based internship/project recommendations**.
- Enabled **centralized allocation and tracking** of internships and projects with ranked recommendations and TPO/faculty approval workflows.

Distributed Rate Limiter **2026**
Technologies: C++17, Redis, Multithreading, Data Structures

- Built a **distributed API rate limiter in C++** using Redis and the token bucket algorithm to control request traffic across multiple server instances.
- Implemented **atomic Redis operations and key expiration** to maintain consistent request limits and automatically reset inactive rate-limit states.
- Added **concurrent request handling** and separated rate-limiting logic from request processing for easier integration with backend services.

Technical Skills

Languages	C, C++, Java, Python, JavaScript
Backend	Node.js, Express.js, FastAPI, REST APIs, JWT, Authentication, Authorization
MERN	MongoDB, Express.js, React, Node.js, Mongoose, HTML, CSS
Databases	MongoDB, MySQL, Redis, SQL, Database Design, CRUD, Indexing
AI / LLM	RAG, Embeddings, FAISS, LLM APIs, Prompt Engineering, LangChain
Python	Type Hints, Clean Code, Async, REST API Development, Testing
CS Fundamentals	Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Tools	Git, GitHub, Postman, VS Code, npm, Docker Basics

Achievements and Certifications

- Solved **700+ coding problems** across multiple platforms, including **450+ on LeetCode (Rating: 1594)**.
- **Winner – Tekathon 2025**, Chandigarh University; secured first place in the **Smart Education** category.
- **Certification on NPTEL – Cloud Computing** and **Full-Stack Development – Angular and Spring Boot.**